

q+a link

<https://tinyurl.com/140zoom>

Lecture 3: Episodic Memory, Part 1

PSYCH/COGS 140M 2026.04.09

Reminder

Quiz 1 - Next Thursday, April 16th

- Will be **entirely online**
- Available via Canvas starting at **12:30pm Thursday** until **Friday 5pm**
 - Once you start, you have three hours
 - Honestly, should not require even a whole hour
- RespondUs browser (no camera 🙅🏻)
- **Question bank** is on Canvas (and has been since last week)
- Please follow all UCI honor codes
- Grades should be posted by the following Thursday (*-ish*) after the quiz

Outline

- Defining and measuring episodic memory
- Encoding
- The medial temporal lobe (MTL) and Amnesia
- Consolidation
- Retrieval

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Multiple memory systems

Transient
memory

Sensory Short-term
(working)

Long-term
memory

Declarative

Non-declarative

Semantic

Episodic

Conditioning

Skill-learning

Priming

Working vs long-term memory

- WM is **transient**, whereas LTM is **durable**
 - Where is the line between the two?
- The difference has less to do with time, and more to do with whether information is *actively* in mind
 - If information is still active, rely upon WM to retrieve it
 - If attention has shifted and information is no longer active, rely upon LTM to retrieve it
 - Still room for theory to formalize this distinction

Declarative memory

- Episodic memory
 - Conscious memories of past events
 - Memory for a specific event (what) that occurred at a specific time (when) and place (where)
- Semantic memory
 - General knowledge about the world, i.e., knowing facts and concepts



EPISODIC & SEMANTIC MEMORY IN
THE GOOD PLACE

Terminology

- Encoding
 - **Forming** a memory representation
- Consolidation
 - **Transforming** an encoded experience into a durable memory
- Retrieval
 - **Bringing** information back to mind

Measuring episodic memory

- How do we measure episodic memory in the lab?
 - Recall
 - Recognition
 - Tests in animals

Recall & Recognition

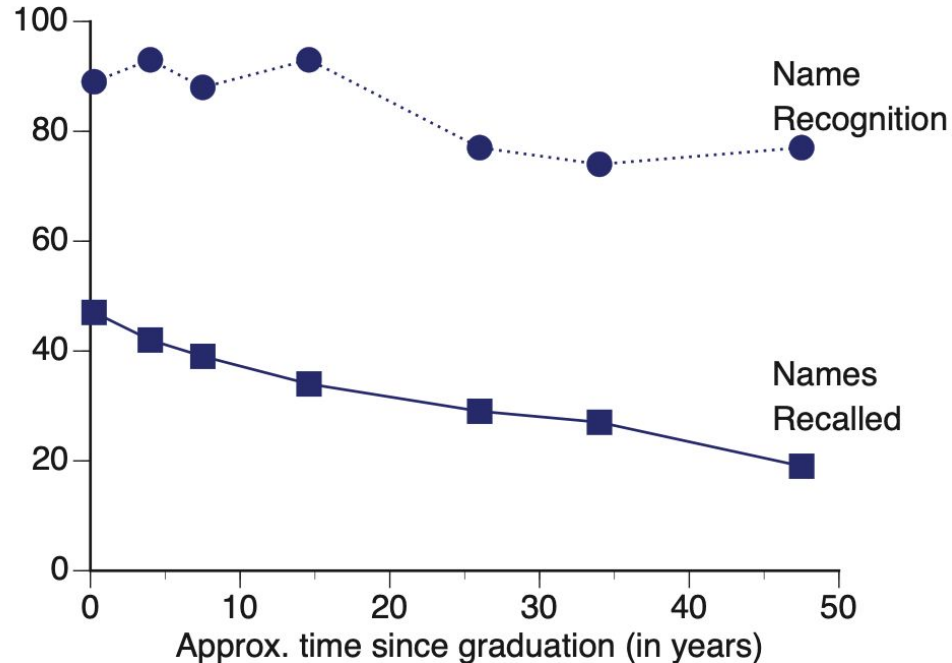
- Free recall
 - Unconstrained retrieval; try to remember a set of information in any order
 - E.g., study a list of words, then recall as many as you can
- Cued recall
 - Try to remember a specific piece of information that matches specific cue or query
 - E.g., study a list of words, then be tested with “ba____,” or “mo_____”
- Recognition
 - Did you see this before? (yes or no)
 - E.g., study a list of words, then make decision about whether “bacon” was on the list

Type of Test Affects

Assessment of Memory

Memory for High School Classmates

Percent Correct



Bairick et al.
(1975)

Type of Test Affects Assessment of Memory

Memory for Word Lists



(Graf, Squire, & Mandler, 1984)

Testing episodic memory in animals

- Do non-human animals have episodic (“declarative”) memory?
- Given that we can’t communicate with them verbally, how can we test this?
 - Do they have memory for what/where/when?

The Amazing Scrub Jay!

Scrub Jays remember:

- Where they buried their food. Evidence: they find it with great precision!
- What they buried. Evidence: they go to the best stuff first
- When they buried it, because they won't retrieve food that is likely to have spoiled



Scrub Jay

If scrub jays remember “what,” “where,” and “when” is that episodic memory?

Episodic Memory

encoding part 1

Outline

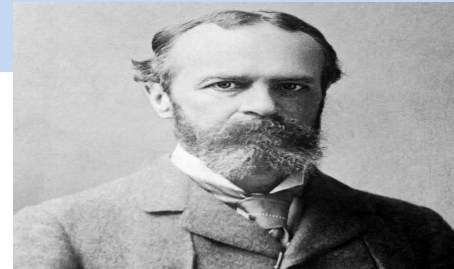
- Defining and measuring episodic memory
- **Encoding**
- The medial temporal lobe (MTL) and Amnesia
- Consolidation
- Retrieval

Factors that Influence Encoding

- Attention
- Repetition
- Spacing
- Retrieval practice
- Depth of processing

Defining attention

“Everybody knows what attention is ... it is the taking possession by the mind, in clear and vivid form, of one out of what seem several simultaneously possible objects or trains of thought. Focalization, concentration, of consciousness are of its essence. It implies withdrawal from some things in order to deal effectively with others.”



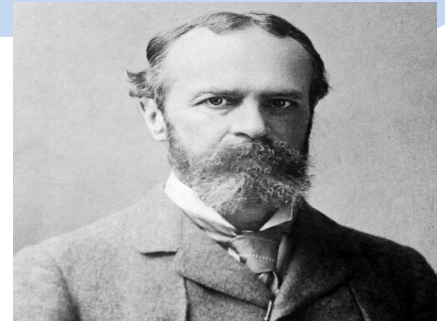
William James, 1893₂₀

Defining attention

- Processes by which we allocate our limited processing resources across competing demands
- Attention is *selective*
 - Ignoring a conversation behind you during lecture
- Attention is *limited*
 - It's challenging to try to read slides, listen to me, and take notes at the same time (divided attention)
 - It's also challenging to try to remain focused on lecture material for 80 minutes (vigilance)

Attention & Memory

“Whatever may be the kind of attention, voluntary or involuntary, it always acts alike; the image of an object or event is capable of revival, and of complete revival, in proportion to the degree of attention with which we have considered the object or event. We put this rule in practice at every moment in ordinary life.”



William James, 1890

Selective attention

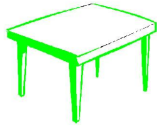
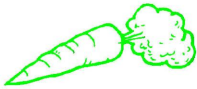
- Involves selecting among multiple streams of sensory information
- Selection, however, causes you to miss information that you don't pay attention to: **inattention blindness**
- Selective attention task (Simons & Chabris, 1999)
- Movie perception test (Levin & Simons, 1997)

Selective attention

- How do we choose to select one stream of information over another?
- Dual attention theory
 - **Top-down** (endogenous): goal-directed shift of attention
 - **Bottom-up** (exogenous): reflexive shift of attention by salient or unexpected stimuli

Distinctiveness, attention, and memory

- Isolation effect
 - Items or events that are distinctive (isolated) are better remembered than typical items or events

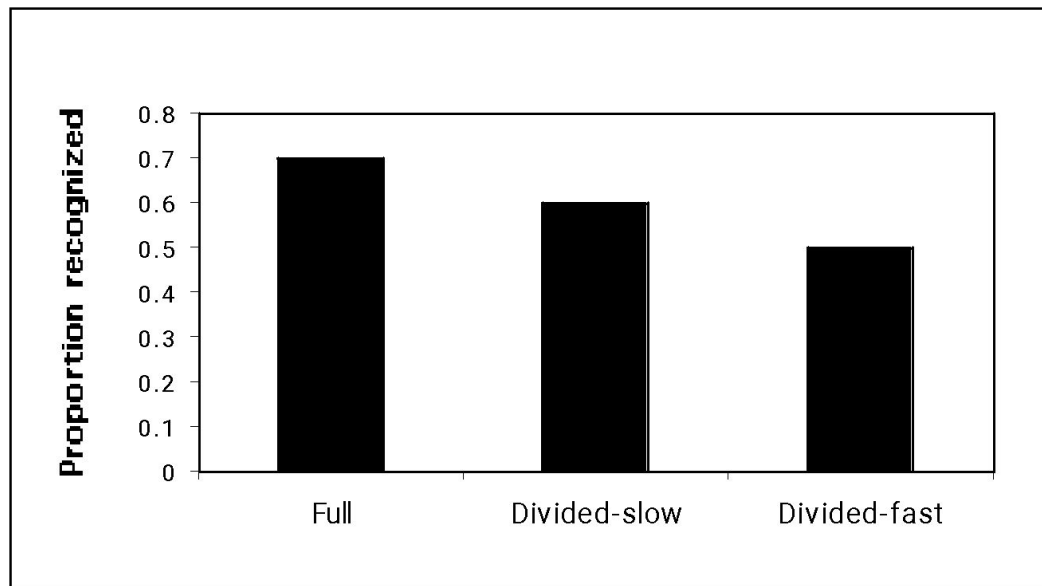


- Effect may be due in part to distinctive items reflexively grabbing your attention (bottom-up)

Divided attention

- Given the importance of attention during learning, what happens when our attention is divided?
- Gardiner and Parkin (1990)
 - Full attention
 - Learned a list of visually-presented words
 - Divided attention
 - Word learning plus a secondary task involving monitoring the pitch of presented tones
 - Rate of tone presentation was either fast or slow

Gardiner and Parkin (1990)



Attention during encoding is critical for effective learning

Divided attention in the real world

How effective do you think your studying is if you simultaneously...

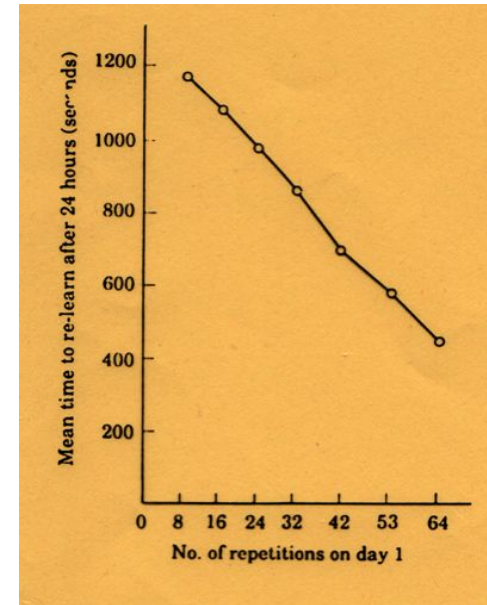
- Watch TV
- Run on the treadmill
- Listen to music
- Eat dinner
- Intermittently receive texts

Factors that Influence Encoding

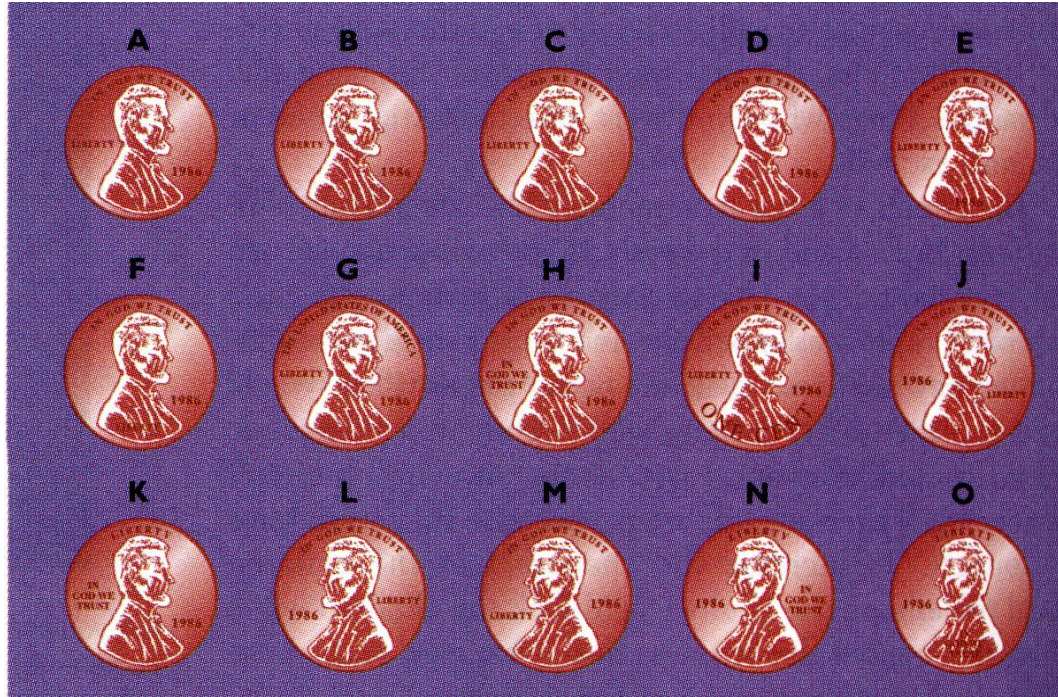
- Attention
- Repetition
- Spacing
- Retrieval practice
- Depth of processing

Ebbinghaus (1850 – 1909)

- Studied lists of nonsense words and assessed the influence of different variables on memory
- Found that repeatedly studying words reduced forgetting
 - The more repetitions, the better his memory performance



Caveats to repetition



Merely being exposed to information repeatedly does *not* ensure that it will be well learned!

Factors that Influence Encoding

- Attention
- Repetition
- **Spacing**
- Retrieval practice
- Depth of processing

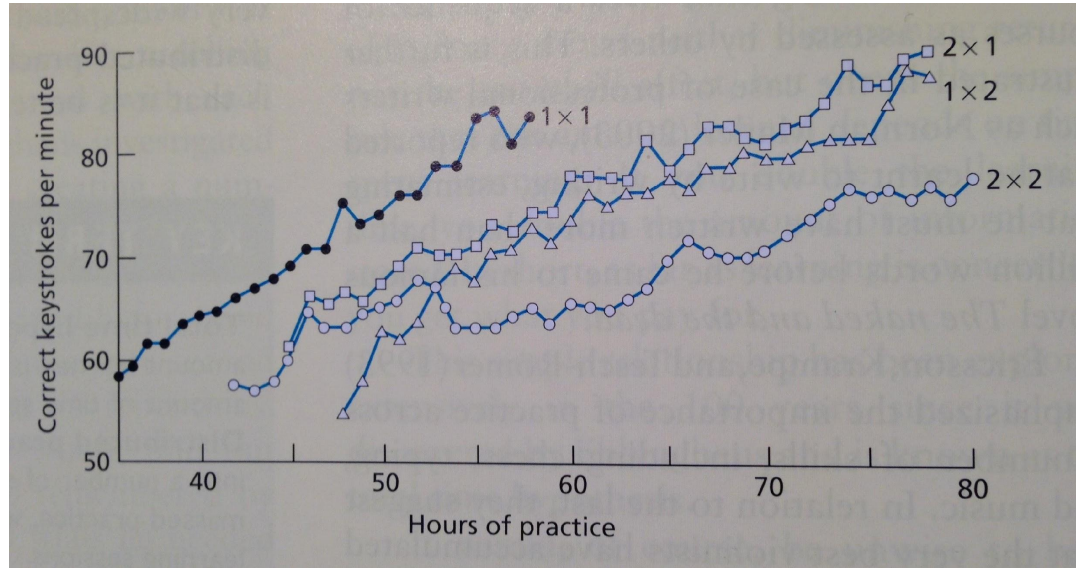
The spacing effect

- Also discovered by Ebbinghaus:
 - “With any considerable number of repetitions a suitable distribution of them over a space of time is decidedly more advantageous than the massing of them at a single time”
- I.e., greater lags between study trials (or practice sessions) lead to better long-term retention
- True for most forms of long-term memory, e.g., semantic learning, skill learning, habituation/sensitization

Baddeley and Longman (1978)

- British post office needed to train a large number of postal workers to type
- Four schedules
 - Two two-hour sessions per day (2x2)
 - One two-hour session per day (1x2)
 - Two one-hour sessions per day (2x1)
 - One one-hour session per day (1x1)

Baddeley and Longman (1978)



- Spaced group (1x1) showed the fastest learning rate, learned in the fewest number of hours, and had the best retention after four months
- But... the 1x1 group was the least content with training!

Blocked vs random practice

- Imagine trying to learn two new skills; how should you distribute your practice to learn most effectively?

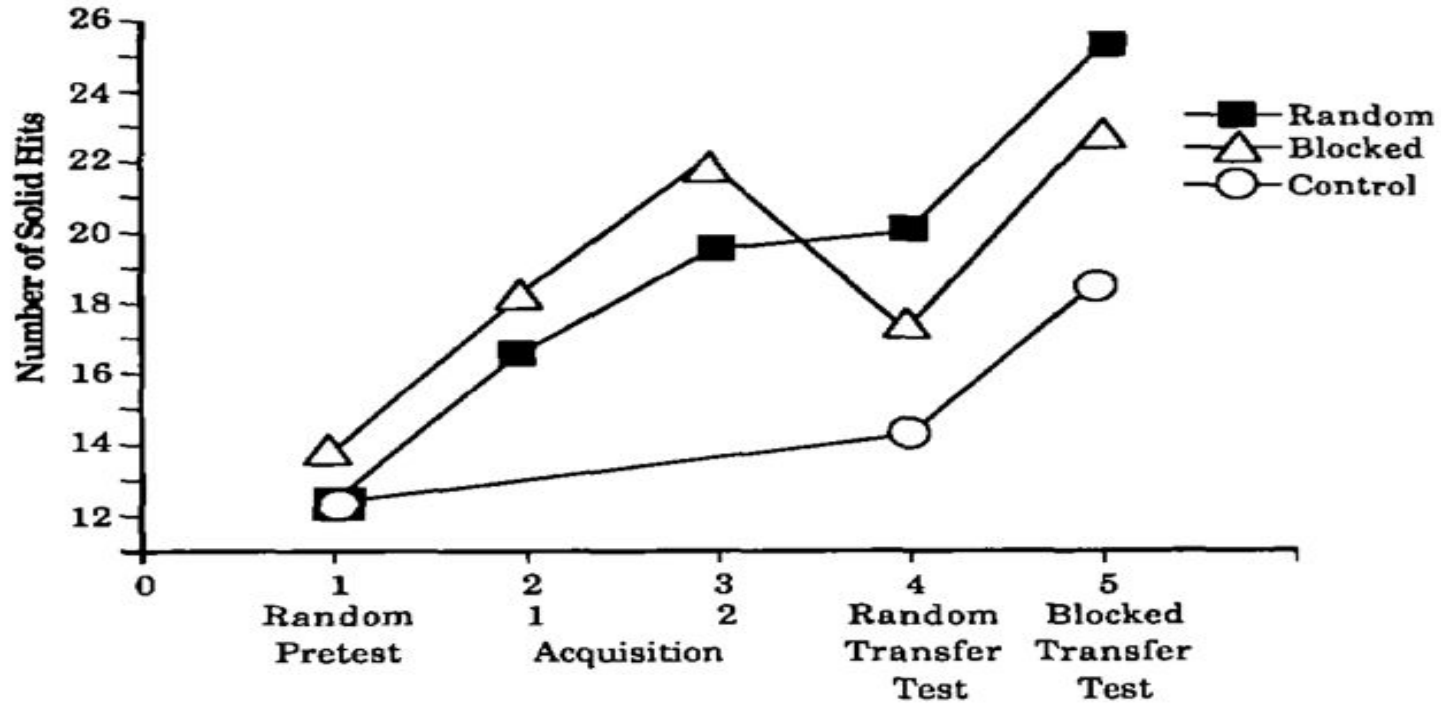
Blocked practice

Skill 1
Skill 1
Skill 1
Skill 1
Skill 2
Skill 2
Skill 2
Skill 2

Random practice

Skill 1
Skill 2
Skill 1
Skill 2
Skill 2
Skill 1
Skill 1
Skill 2

Hull et al (1994)



Interpreting the spacing effect

- Deficient processing hypothesis

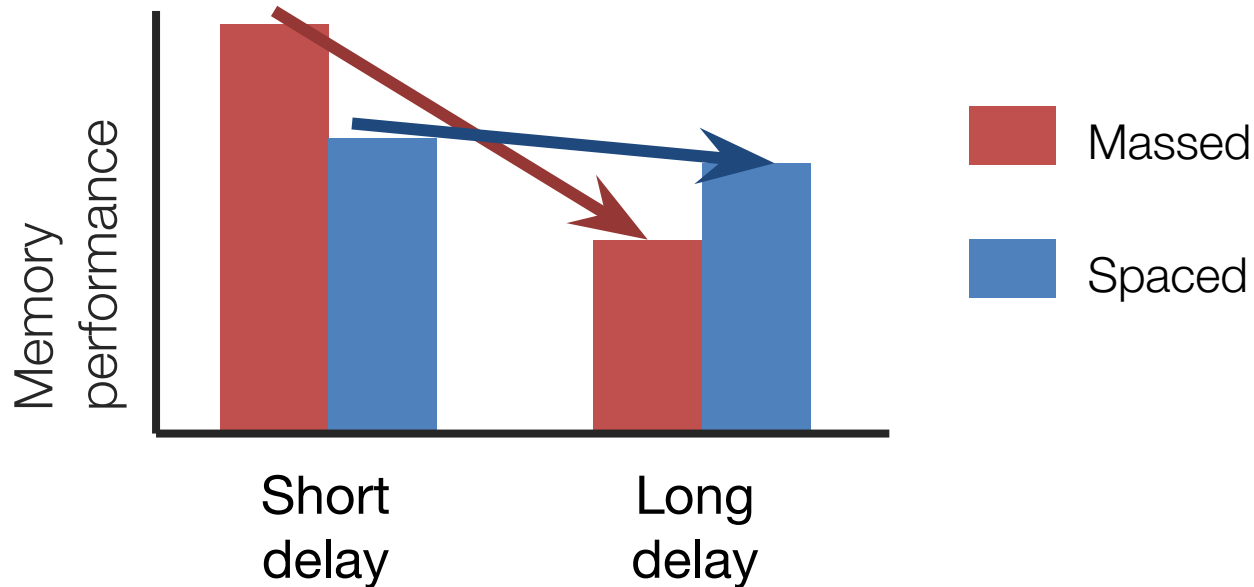
- During massed training, repeated occurrences of an item are not processed fully (reduced attention)

- Encoding variability hypothesis

- Spaced training leads to more variable encoding (e.g., different encoding contexts)
- Variable encoding allows for better retrieval because we can rely on multiple retrieval cues

Caveats to the spacing effect

- Spaced training is more effective than massed training, but typically only at long delays



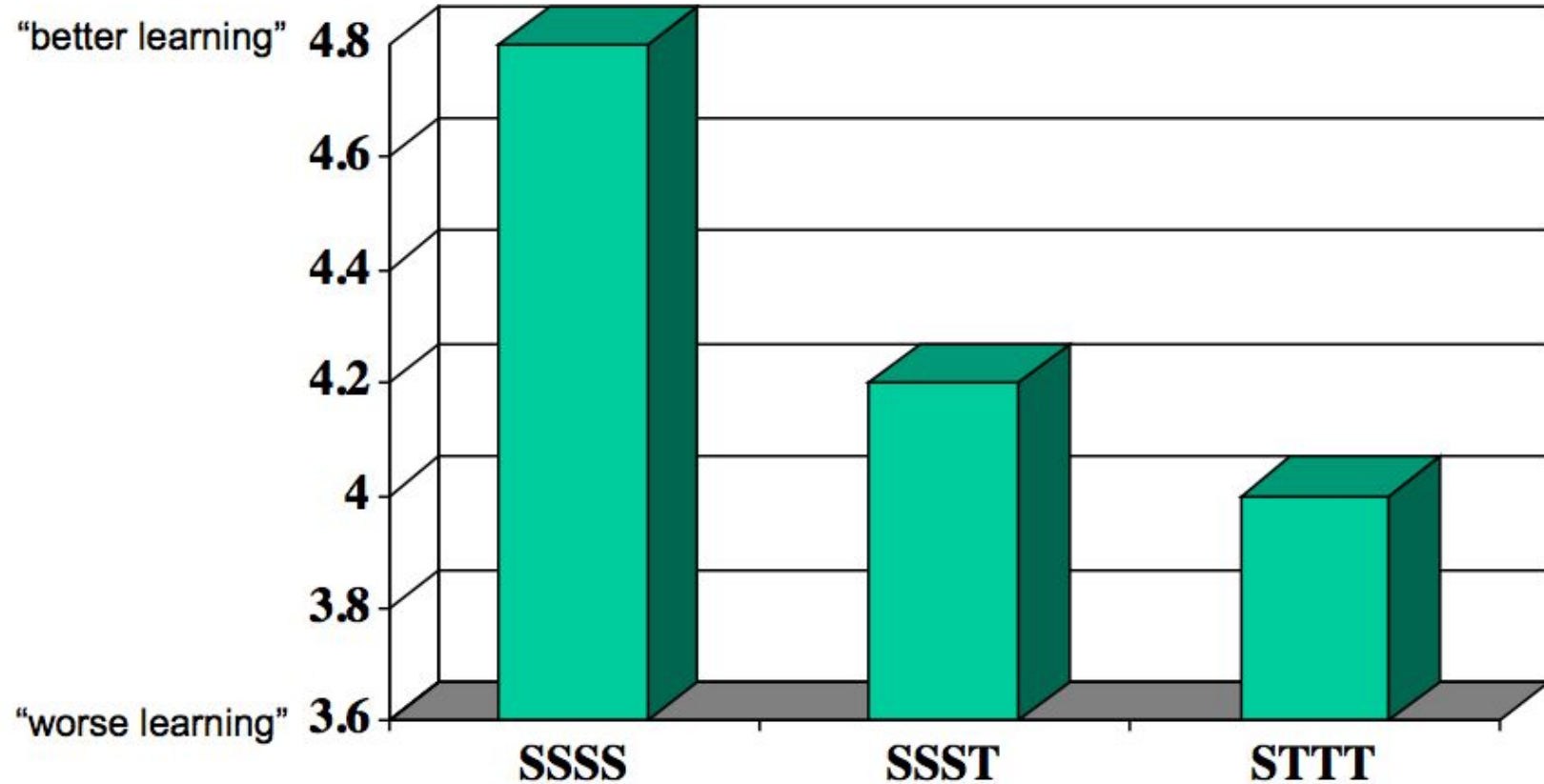
Factors that Influence Encoding

- Attention
- Repetition
- Spacing
- **Retrieval practice**
- Depth of processing

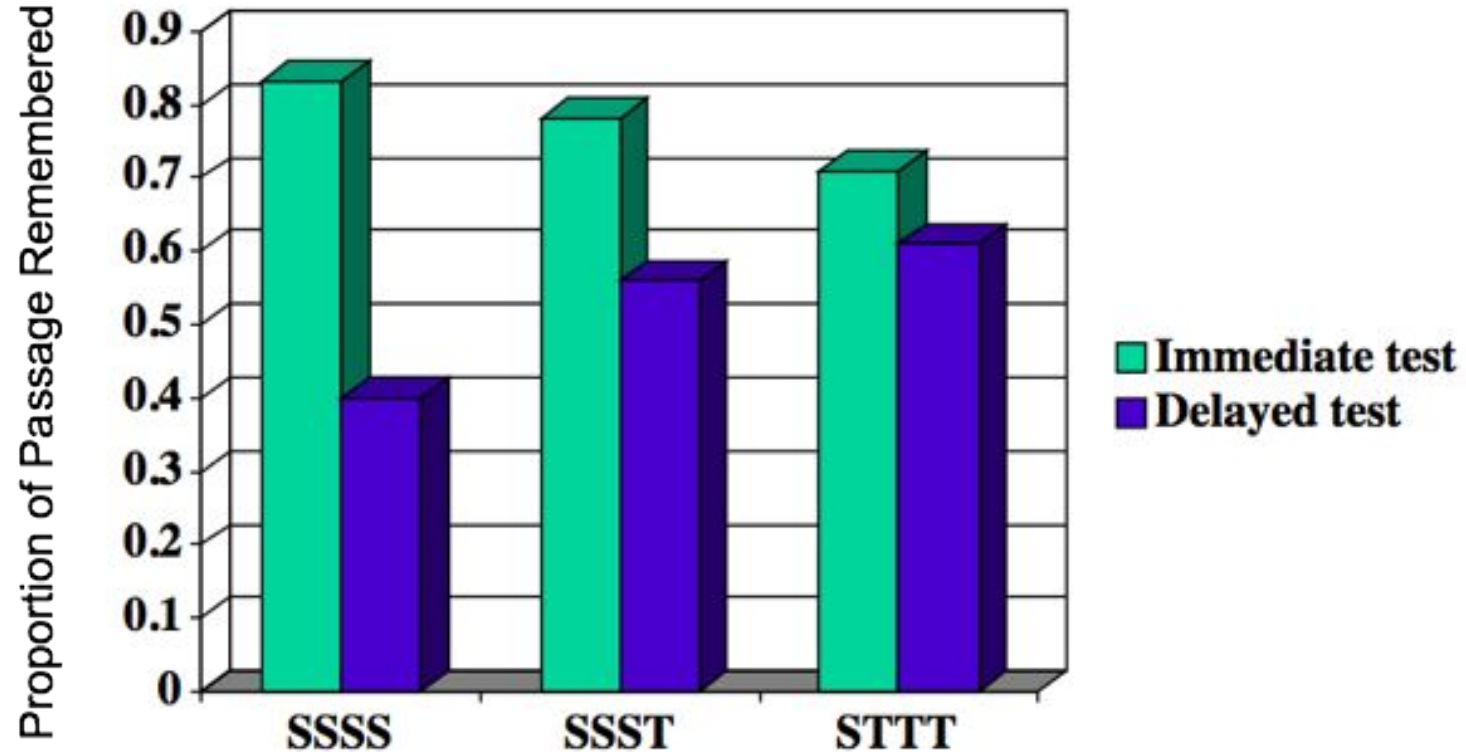
Roediger and Karpicke (2006)

- Participants studied a passage of text
- Assigned to one of three strategies:
 - Four study episodes (SSSS)
 - Three study episodes and one test (SSST)
 - One study episode and three tests (STTT)
- Asked to rate how well they felt that they learned the material—judgment of learning
- Tested either 5 minutes or 1 week later

Judgments of learning (JOL)



Memory performance



Caveats to retrieval practice

- Feedback is important
- Errors can disrupt later performance
- Advantages of retrieval practice are most evident at longer delays

Real-world examples

- Which class schedule will lead to better performance on the final exam? (+ Which schedule will students like better?)

Lecture
Review
Lecture
Review
Lecture
Review
Lecture
Review
Lecture

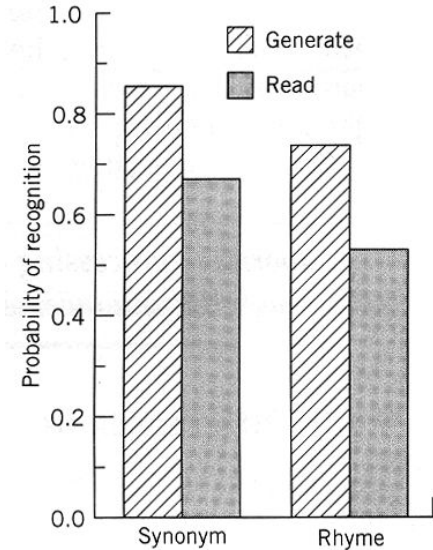
Final Exam

Lecture
Quiz
Lecture
Midterm
Lecture
Quiz
Lecture
Midterm
Lecture
Quiz
Lecture

Final Exam

Generation effect

- You are more likely to remember material if you generate it yourself, rather than simply being exposed to it
- Read condition
 - Synonym: Unhappy: Sad
 - Rhyme: Lad: Sad
- Generate condition
 - Synonym: Unhappy: S_____
 - Rhyme: Lad: S_____

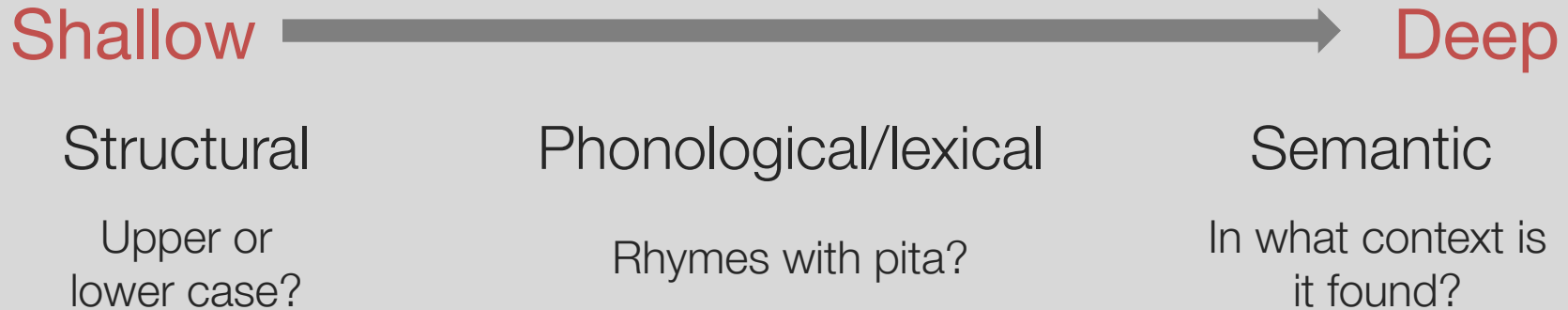


Outline

- Description of forgetting project
- Attention
- Repetition
- Spacing
- Retrieval practice
- Depth of processing

Depth of processing

- Stimuli and events can be processed at different levels
 - Deeper levels of processing lead to better memory



Understanding depth of processing

- Levels of processing differ with respect to:
 - Which features of an item/event are encoded
 - The degree to which you relate new information to prior knowledge
- Semantic (deep) processing may be more distinctive, leading to less interference across items
- Semantic processing links new information to existing knowledge, providing more retrieval cues at test

Caveats to depth to processing

Transfer Appropriate Processing (TAP)

Morris et al. (1977)

Subjects performed one of two study tasks

- In each case, they had to say whether a target word fit into the blank
- In one condition, this was based on meaning
 - The _____ was on the shelf
 - “book” - yes or no
- In another condition, this was based on sound
 - _____ rhymes with fear
 - “spear” - yes or no

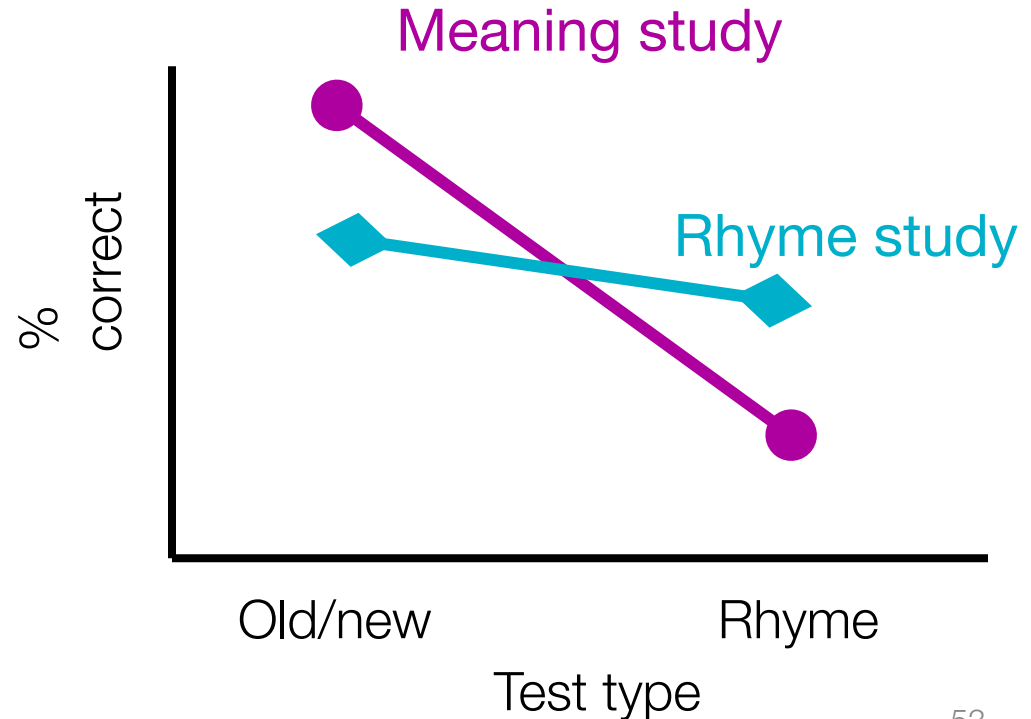
They were then tested using two different types of test

- Standard recognition test (“Did you see ‘book’ before?”)
- Rhyme test (“Did you see a word that rhymes with ‘clear’ before?”)

Caveats to depth to processing

TAP: Morris et al. (1977) study

- Recognition (old/new) performance was better following meaning task
- Rhyme performance was better following rhyming task
- Deep study is not always better



Intuitions about Learning

Which of the following is the MOST important ingredient for successful learning?

1. The intention and desire to learn
2. Paying close attention to the material as you study
3. Learning in a way that matches your personal learning style
4. The amount of time you spend studying
5. What you think about while studying

Intuitions about learning

- Strategies that we think are helpful often are *not* very effective
- We often use strategies that maximize short-term retention but lead to poor long-term retention (e.g., cramming)
- The best strategies for learning are often ones that we like least!
 - Desirable difficulties

Does effort or intention to learn matter?



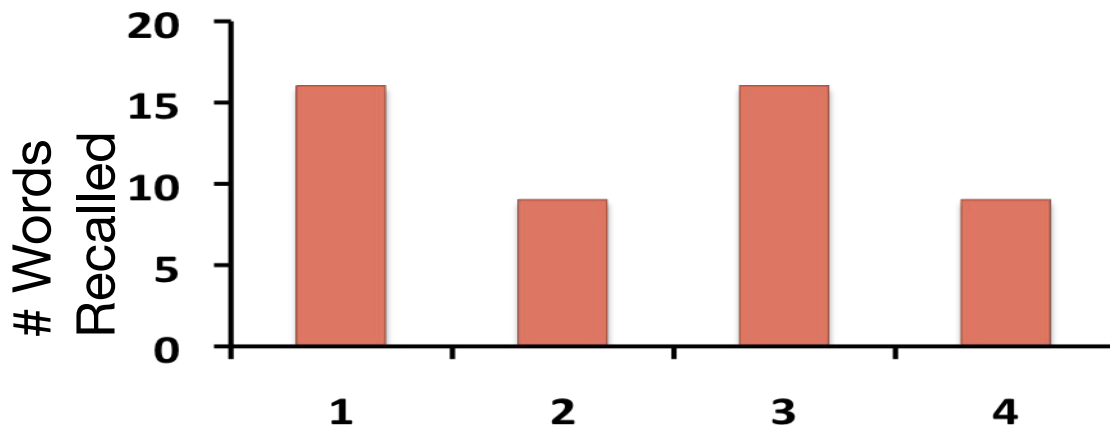
Intending to Learn

- Orienting task
 - Deep: rate pleasantness of words
 - Shallow: does word contain a “Q” or an “A”?
- Learning purpose
 - Intentional: forewarned about memory test
 - Incidental: didn’t know a memory test was coming

	Deep	Shallow
Incidental		
Intentional		

Intending to Learn

Hypothesis #1: Intent to learn is critical

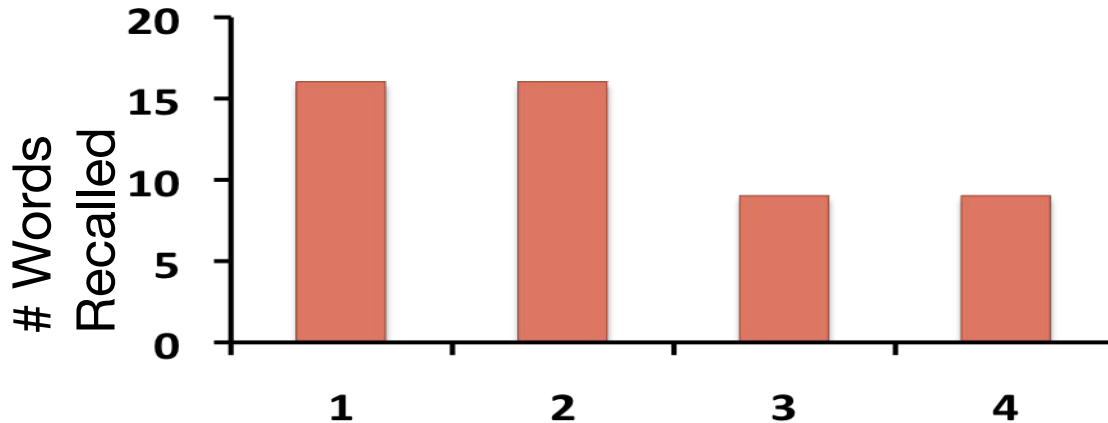


Four Groups:

1. Deep/**I**ntentional
2. Deep/**I**ncidental
3. Shallow/**I**ntentional
4. Shallow/**I**ncidental

Intending to Learn

Hypothesis #2: Depth of processing is critical

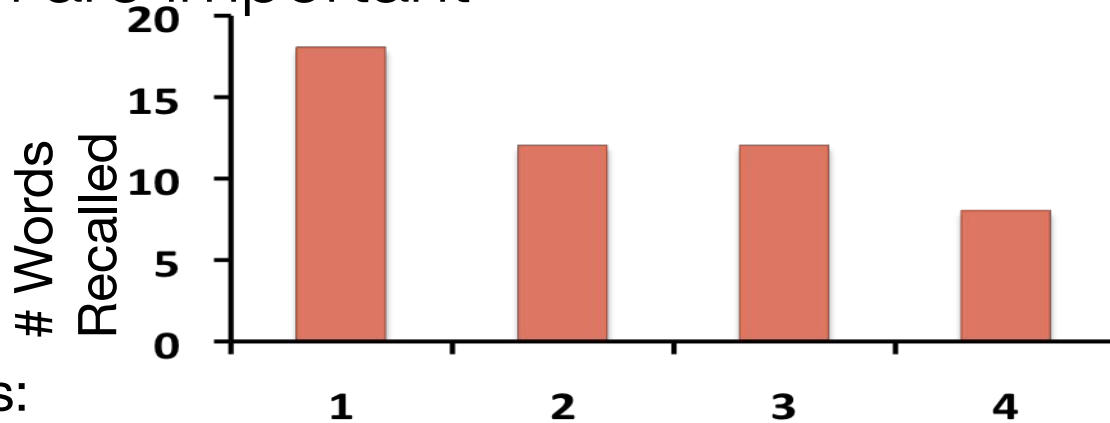


Four Groups:

1. Deep/Intentional
2. Deep/Incidental
3. Shallow/Intentional
4. Shallow/Incidental

Intending to Learn

Hypothesis #3: Both depth of processing and intent to learn are important



Four Groups:

1. Deep/Intentional
2. Deep/Incidental
3. Shallow/Intentional
4. Shallow/Incidental

Intending to Learn

Words Recalled as a Function of Orienting Task and Participant Awareness of Learning Task

Learning-Purpose Conditions	Words Recalled (%) Orienting Task	
	Rate	
	Pleasantness	Check Letters
Incidental	68	39
Intentional	69	43

After Hyde and Jenkins, 1973. Adapted by permission of the publisher.
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Next up

Amnesia - don't forget!